National Geospatial-Intelligence Agency (NGA)

Foundation Digital Twin (FDT)
Automated Feature Extraction (AFE)

Statement of Objectives (SOO)
Version 1.0

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CHANGE HISTORY

Date	Section Changed	Reason for Change	Author	Revision
2/25/26		Initial Issue/RFI Release		1.0

1.0 Summary and Background

To overcome a historically fragmented data landscape where critical information is trapped in disconnected, stove-piped systems, the Foundation Digital Twin (FDT) program requires an Automated Feature Extraction (AFE) capability. This automated engine is essential for autonomously scanning vast, disparate datasets to identify and catalog the critical infrastructure and objects of interest that form the core of its living, authoritative digital replica of the globe. By pivoting from an outdated, product-focused model to this unified, data-centric architecture, the AFE process is an essential component for achieving true data harmonization and maintaining a temporally relevant and spatially accurate version of Foundation GEOINT for users.

The overall purpose of this Statement of Objectives is to acquire a contractor who can provide robust Automated Feature Extraction (AFE) solutions. To respond, vendors are required to select features from the list in Appendix A and, for each selected feature, provide a detailed explanation of how their solution meets the requirements of the specific use cases described in Section 3.0. These solutions will modernize NGA's geospatial intelligence production by leveraging AI/ML to automate manual processes, improve data accuracy, and ensure seamless integration with critical NGA production systems. The successful implementation of AFE capabilities will significantly enhance timeliness, relevance, and accuracy of geospatial intelligence, directly supporting national security missions.

The National Geospatial-Intelligence Agency (NGA) is a Department of War (DoW) combat support agency and a member of the National Intelligence Community (IC). NGA develops imagery and map-based intelligence solutions for U.S. national defense, homeland security, and safety of navigation. NGA's mission is to "provide timely, relevant, and accurate geospatial intelligence in support of national security."

The current Foundation GEOINT environment faces significant operational challenges that impede efficiency and effectiveness. Foundation GEOINT (FG) Analysts spend extensive time on repetitive manual feature extraction, conflation, and validation, which are time-consuming activities that could be automated. To address these challenges and support NGA's 2035 GEOINT Concept of Operations, NGA seeks to implement AFE capabilities. These solutions should leverage artificial intelligence and machine learning (AI/ML) to transform geospatial intelligence production, significantly reducing manual processing time, increasing data accuracy, and enabling the rapid delivery of intelligence to decision-makers.

2.0 Scope of Work

The scope of this effort is to acquire services or capabilities for AFE capabilities focused on features identified by FG as important but challenging. This includes the provisioning, configuration, deployment, and sustainment of a commercial/out-of-the-box AFE capability to automate the identification and extraction of geospatial features from imagery and other source materials. AFE capabilities will be a critical enabler for modernizing Foundation GEOINT processes and will be integrated with key NGA systems. The work will support NGA's maritime, aeronautical, topographic, geographic, and geomatics mission domains and will be integrated into NGA's FG production environments on all required security domains.

3.0 Objectives

The contractor shall achieve the following four high-level objectives to successfully deliver a comprehensive Automated Feature Extraction (AFE) solution:

Objective 1

Provide and Deploy a High-Performing Automated Feature Extraction (AFE) Capability as a Service

The primary goal is to provide, deliver, and sustain a robust AFE capability that automatically detects and extracts geospatial features from various sources including imagery (EO, SAR, HS, MS) and raster maps. This system must be capable of monoscopic and/or stereoscopic extraction and handle individual and batch processing. The capability shall automatically attribute extracted features based on observable characteristics in accordance with NGA domain standards and preserve feature geometry, spatial relationships, and metadata from the source data.

Objective 2

Ensure High-Quality Data and Enable Analyst-in-the-Loop Validation

The contractor shall implement a system that ensures the accuracy and reliability of all extracted data. This includes achieving a threshold accuracy of 80% for topographic and maritime features and 90% for aeronautical features, with an objective accuracy of 99%. The system must feature automated quality assessment processes with confidence scores and provide an intuitive interface for analysts to review, validate, and correct extracted features. All analyst feedback shall be used to continuously improve the performance of the AI/ML models.

Objective 3

Enable Seamless Utilization by NGA's GEOINT Ecosystem via Open APIs

FG desires that AFE capabilities be exposed as a service that is callable and accessible from NGA Architecture via open APIs. This requires seamless interoperability with the systems and data repositories responsible for geospatial AFE data generation and AFE repository management. The contractor's solution shall provide a documented set of open APIs and utilize NGA standardized formats (e.g., GDB, Shapefile) for all extracted features to ensure seamless interoperability with current enterprise systems and adaptability for any future integrations.

Objective 4

Meet and Maintain Stringent Performance and Security Standards

The solution must be scalable, reliable, and secure. The system shall be capable of processing a standard 1GB source file in less than 10 minutes and achieve 99% availability in the production environment. The contractor must ensure the system is deployable to both Unclassified and Classified domains and complies with all applicable security directives, including DNI ICD 503 and all requirements outlined in DD Form 254.

Appendix A: Prioritized Geography Features for AFE

Automated Feature Extraction Features of Interest:

Automated feature extraction consisting of features represented by points, lines, and polygons. Collection of topographic features is global. Full specifications describing these features and their associated attributes are available for each data program and should be consulted in the development of AFE processes.

Generally, features are collected to the most complex geometry visible at the scale of collection.

Prioritized Features consist of the following list below. This list is an amalgam of TDS and MUVD collectable features and provides a representative sample of the feature types collected in these topographic data programs. Note that some feature types may lump together significant and important categories of interest that are typically distinguished by attributes (e.g., the Feature Function attribute on a Building feature determines if a Building is a hospital or a religious building which are important categories for the topographic mission):

Prioritized features for AFE:

Apron
Bridge
Building
Cable
Canal
Cart track
Cistern
Culvert
Dam
Ditch
Electric power station
Elevation contour
Extraction mine
Fence
Ford
Gate
Helipad
Hut
Inland waterbody
Island
Military installation
Non-water well
Pipeline
Power substation
Pylons
Railway
River
Road
Runway
Shoreline
Sports ground
Storage Tank
Taxiway
Tower
Trail
Tunnel
Wall
Water tower
Water well

DEFINITIONS OF PRIORITIZED FEATURES:

Apron

A defined area, on a land aerodrome/heliport, intended to accommodate aircraft/helicopters for purposes of loading and unloading passengers, mail or cargo, and for fueling, parking or maintenance.

Bridge

A structure that connects two locations and provides for the passage of a transportation route (for example: a road or a railway) over a terrain obstacle (for example: a waterbody, a gully, and/or a road). A bridge consists of a set of two abutments and/or zero or more bridge piers joined by bridge spans. A bridge may serve, for example, as an overpass or a viaduct. In the context of a bridge, the scope of the term 'transportation route' includes the transportation of liquids or gases by means of either pipelines or aqueducts.

Building

A free-standing self-supporting construction that is roofed, usually walled, and is intended for human occupancy (for example: a place of work or recreation) and/or habitation. For example, a dormitory, a bank, and a restaurant. Of particular interest is the ability to identify those buildings that function as hospitals/medical facilities, schools/educational facilities, and places of worship.

Cable

A single continuous rope-like bundle consisting of multiple strands. The strands may be individually insulated and/or protected and the cable as a whole sheathed. Cables may be used for load bearing (for example, supporting or suspending equipment and/or structures), transmitting electrical power, and/or communicating signals (for example, by electrical or optical means).

Canal

An artificial waterway with no flow, or a controlled flow, usable or built for navigation.

Cart track

An unimproved road. The surface is usually rough (for example: rutted) and minimally prepared (for example: packed earth or thinly covered with gravel). An unimproved, natural travelled way that can accommodate four-wheeled or tracked vehicles and carts.

Cistern

A man-made container used for the collection and/or storage of water.

Culvert

An enclosed channel for carrying a watercourse (for example: a stream, a sewer, or a drain) under a route (for example: a road, a railway, or an embankment). Usually, the construction of the route is unaffected.

Dam

A barrier constructed to hold back water and raise its level to form a reservoir or to prevent flooding.

Ditch

An artificial waterway with no flow, or a controlled flow, usually unlined, used for draining or irrigating land.

Electric power station

A facility including one or more buildings and equipment used for electric power generation. An electric power station consists of one or more power generating units, each consisting of the full set of equipment required to generate power and capable of independent operation. The power generating units are located on one or more contiguous or adjacent properties, are under the common control of the same entity and supply power through a common connection to the electric grid. Electric power stations most commonly are used to generate electricity for long distance transmission.

Elevation contour

An elevation contour is a line that connects points of the same elevation.

Extraction mine

An excavation made in the terrain for the purpose of extracting and/or exploiting natural resources.

Fence

A man-made barrier of relatively light structure used as an enclosure or boundary. Similar structures that are constructed of heavy materials (for example: stone, rock or masonry) are classified as walls.

Ford

A shallow place in a body of water used as a crossing. A Ford is a location in a body of water where the physical characteristics of the current, bottom materials, and approaches for a road, cart track, or trail permit the passage of personnel and/or vehicles.

Gate

A barrier on a transportation route (for example: a road, a railway, a tunnel, or a bridge) that controls passage (may be opened and closed).

Helipad

A designated area, usually with a prepared surface, used for the take-off, landing, or parking of helicopters. This prepared surface could either be located on land or on a platform over water. It may or may not be associated with an aerodrome. For example: a hospital helipad, and an offshore rig helipad.

Hut

A small, simple free-standing (detached) self-contained residence usually having only a single multi-function room.

Inland waterbody

A body of water that is entirely surrounded by land. It may occur in a natural terrain depression in which water collects, or may be impounded by a dam, or formed by its bed being hollowed out of the soil, or formed by embanking and/or damming up a natural hollow (for example: by a beaver dam). Inland waterbodies have many uses such as: a source of water for irrigation, industrial processes, human consumption, and recreation. Impounded inland waterbodies may also be used for flood control. The most common Inland Waterbody feature types are the: Lake, Pond and Reservoir.

Island

A land mass, other than a continent, surrounded by water.

Military installation

An installation designed for military use. For example, used to perform military operations, initiate forward movements, and/or furnish supplies. Often protected by fortifications or natural advantages.

Non-water well

A shaft sunk into the ground to reach and tap a supply of liquids and/or gases other than water intended for use in agriculture or domestic consumption. Typically drilled to tap underground reservoirs of hydrocarbons (for example: petroleum or natural gas). May also, for example, yield geothermally heated liquids for use in power generation or heating, or brine for use in the extraction of salt.

Pipeline

A connected set of pipes for conveying liquids, slurries, or gases. Usually for long distances and often located underground.

Power substation

A facility, along a power transmission line, in which electric current is switched, transformed, and/or converted.

Pylons

A pylon or pole used to support one or more cables.

Railway

One or more railway tracks comprising a network that is operated for the conveyance of passengers and/or goods.

River

A natural flowing watercourse, usually meandering through a channel following the lowest topographic valley.

Road

A route with a specially prepared surface that is intended for use by wheeled vehicles.

Runway

A defined rectangular area on a land aerodrome prepared for the landing and take-off of aircraft.

Shoreline

The line where a land mass is in contact with a body of water and the tide state or river stage are unspecified. It may be in either the littoral or inland waters. In the Topographic Data Schema V7.1 this is the feature type Land Water Boundary.

Sports ground

An open area where sporting events, exercises, and/or games occur. For example, an athletic field, a playing field, and/or a sports field.

Storage Tank

A container used for the storage of liquids and/or gases that is not supported by a tower.

Taxiway

A defined path at an aerodrome established for the taxiing of aircraft and intended to provide a ground movement link between one part of the aerodrome and another.

Tower

A relatively tall, narrow structure that may either stand alone or may form part of another structure. Usually of a square, circular, or rectangular cross-section.

Trail

A path worn by the passage of people or animals. Generally unimproved footpaths or animal travel routes that are not sufficiently wide enough to accommodate four-wheeled vehicles or carts.

Tunnel

An underground passage that is open at both ends and usually contains a land transportation route (for example: a road and/or a railway). Commonly used to pass through a hill or mountain, or under a river or road. May also provide underground passage in a mine.

Wall

A solid man-made barrier of generally heavy material used as an enclosure, boundary, or for protection.

Water tower

A tower supporting an elevated storage tank of water.

Water well

A shaft sunk into the ground to reach and tap a supply of water intended for uses other than power generation, heating or the extraction of minerals. May be, for example, drilled to tap deep underground reservoirs or dug to reach a shallow water table. Dug wells are typically circular, lined with masonry, have a stone border and a structure built above then for lowering and raising a bucket.